STAT 361 Assignment 1 Question 1

Edward Tanurkov

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a) Random number generation

```
A <- matrix(round(rnorm(24, mean = 0, sd = 1), digits = 1), nrow = 4, ncol = 6)
A
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6]
## [1,]  1.2  0.6 -0.8 -0.1  0.3 -1.1
## [2,]  1.1  0.8  0.1  1.9  0.4  1.0
## [3,]  0.3 -0.2 -1.6  0.9  1.4  1.6
## [4,] -2.2 -1.1 -1.3 -0.3 -1.0  0.3
```

b) Linear system solution

```
A1 <- A[, -c(2, 5)]
Y = c(1:4)
X = solve(A1)%*%Y
X
```

```
##           [,1]
## [1,] -1.212654
## [2,] -1.843924
## [3,]  2.437281
## [4,] -1.112522
```

c) Solution validation

```
A1X = A1%*%X
same = isTRUE(all.equal(as.vector(A1X), Y))
print(A1X)
```

```
##      [,1]
## [1,]    1
## [2,]    2
## [3,]    3
## [4,]    4
```

```
print(same)
```

```
## [1] TRUE
```

Yes, the solutions are correct since calculating Y via

$$A_1 X$$

is equivalent to

$$Y$$

checked by `iSTRUE(all.equal)`.